Lecture 7: LLM Reasoning

Kai-Wei Chang
CS @ UCLA
kw@kwchang.net

Credit: some slides are modified from Vivek Srikumar & Jason Eisner, Chris Manning

Announcement

- Quiz 2 (Grade scope):
  Will be due today (not grading)
- Assignment 1 (BruinLearn)

Presentation

- Peer review
- Score 1 – 5 (type number)

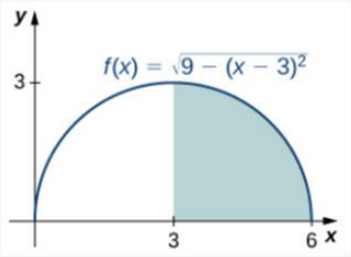
Go to

join.iClicker.com

LCKW

Reasoning

Many Tasks Required Reasoning



ID: 926

Question: What is the size of the shaded area under the curve? Round the answer to 2 decimal places.

Ground truth: 7.07

GPT-4V:
To find the size of the shaded area under the curve $f(x) = \sqrt{9-(x-3)^2}$, we need to integrate the function over the given interval.

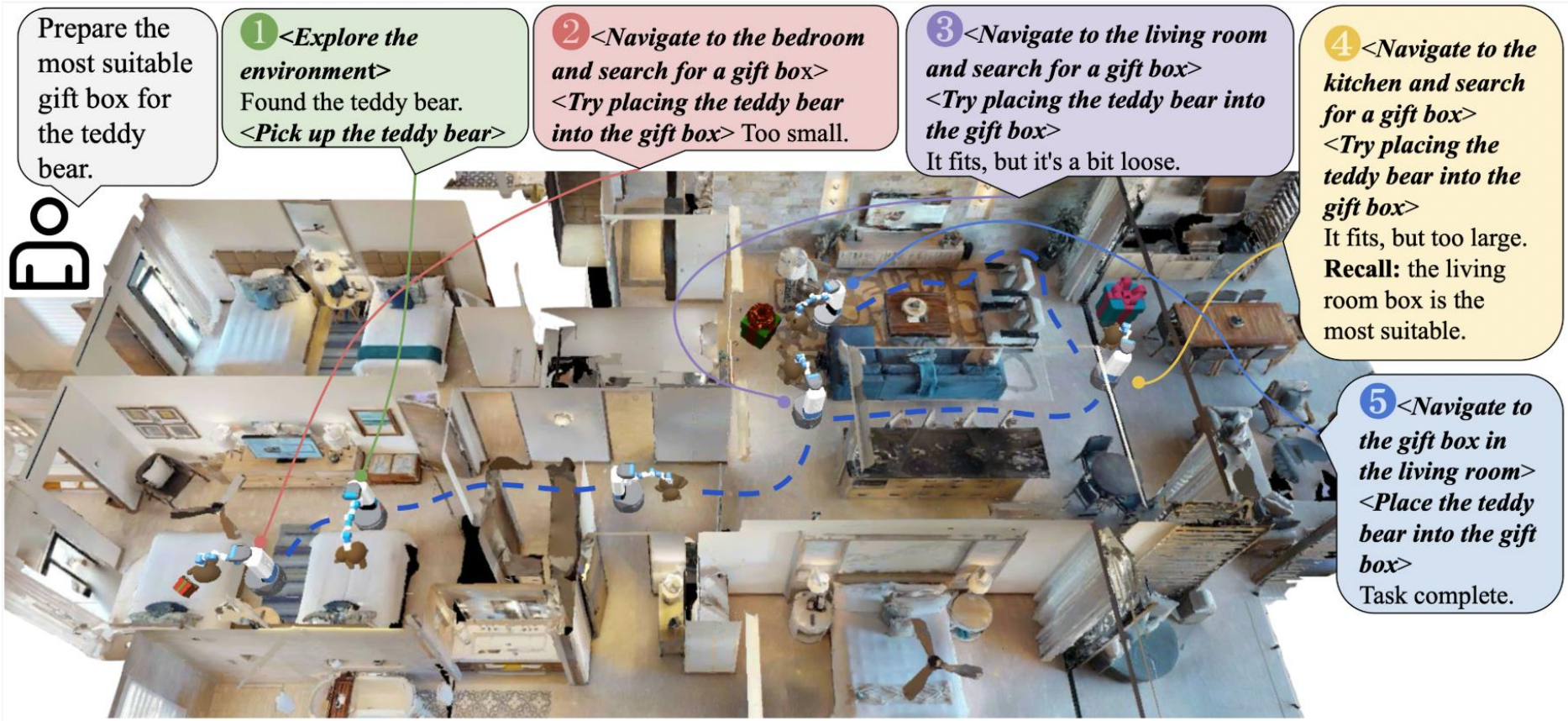
The given function represents the upper half of a circle with a radius of 3 and centered at the point (3,0). The shaded region corresponds to the area of a quarter circle with a radius of 3. The formula for the area of a circle is πr^2. For a quarter of the circle, the area is $\frac{1}{4}\pi r^2$.

Given: $r = 3$

$$\text{Area} = \frac{1}{4}\pi(3^2)$$
$$\text{Area} = \frac{1}{4}\pi(9)$$
$$\text{Area} = \frac{9}{4}\pi$$
$$\text{Area} = 7.07 \text{ (rounded to 2 decimal places)}$$

So, the size of the shaded area under the curve is approximately 7.07 square units.
Correct output: 7.07

Embodied 3D Agentic AI



[3DLLM-Mem: Long-Term Spatial-Temporal Memory for Embodied 3D Large Language Model]

Wenbo Hu, Yining Hong, Yanjun Wang, Leison Gao, Zibu Wei, Xingcheng Yao, Nanyun Peng, Yonatan Bitton, Idan Szpektor, and Kai-Wei Chang, in *NeurIPS*, 2025.

Long-term Memory Embodied Tasks

In-domain Tasks

Instruction: Find an ideal container to store all the cookie dough.

Instruction: Preparing a simple breakfast. The breakfast includes toast with butter and jam, and a cup of coffee.

I need to find a toaster, coffee machine, butter, jam, bread, coffee powder……

No coffee powder found anywhere…
From my memory, the tea bag is on the counter and water kettle on the table.
<Navigate to kitchen>
<Pick up tea bag>
<Put tea bag into the mug>
……
<Navigate to the dining room>
Serve the hot tea instead. Task complete.

+ In-the-wild Challenges

Fine-grained comparison

Force Finish

Task Reschedule

In-the-wild Tasks

Instruction: Collect all cookie dough and then **rearrange** them on the table of living room in descending order of size.

Instruction: **Stop exploring additional rooms** and serve the breakfast now.

I was searching for the **toaster** but I **remembered seeing** the **microwave** in the kitchen. <Navigate to kitchen>

Long-term Memory EQA

Object Counting

Q: How many ceiling lamps are there *after the task finished*?
A: There are 8 ceiling lamps in total.

Comparative

Q: *Which carpet has larger area now*, carpet(0) in room 0 or carpet(1) in room 4?
A: Carpet(0) in room 0 is larger.

Spatial Relation

Q:*After task execution*, if you're standing at table(0), facing chair(1), is the dresser(0) to your left, right, behind?
A:The dresser(0) is to your right.

Object Navigation

Q: If you stand next to the plant(0) facing window(0) in room 15, how to navigate to kitchen cabinet(0)?
A: You need to first turn left and go straight to see stovetop(0), tray(0) and then the kitchen cabinet(0) is right on your front.

Room Layout

Q: In room 8, what is the relative layout of the refrigerator(0), dining table(0), and stove(0) *before task execution*?
A: The refrigerator(0) is located on one side, dining table(0) in the middle, and the stove(0) on the other side of the kitchen.

Captioning

Episodic Memory Captioning

Q: Give a caption covering the summary of the common features across previous exploration and the difference that matters for your current observation.
A: The previous rooms contained storage cabinets, beds, dressers, and kitchen counters ……
This room uniquely features a wardrobe extending along an entire wall and does not contain any pictures, lamps …….

What's reasoning

- Ability to produce correct answers by combining information through multi-steps
- Manipulate intermediate representations to derive better answer
- Maintain consistency across several constraints
- Increase test-time compute to improve performance on a task.

A Brief History of Reasoning

- Latent output representation (Statics)
  - Through graphical models, latent variables (e.g., HMM, Bayesian networks)
  - Through neural network (e.g., hidden layers)
- Latent output representation (Dynamics)
  - Through latent intermediate structures (e.g., parse tree, modules)
  - Dynamically assemble a network or a latent graph per input question
  - E.g., recursive neural network (2013)

- Explicit natural language reasoning (~2017)
  - Natural language explanations for step-by-step solution
  - Required supervision and task-specific design
  - Show significant improvement on algebraic word problems

Learning to Solve and Explain Algebraic Word Problems. ACL 2017.

LLM reasoning

- Prompt Reasoning (~2022)
  - Chain-of-thought
  - Few-shot: provide some reasoning examples <Q> <step-by-step reasoning> <Answer>
  - Zero-shot: simply add “Let’s think step by step.” in the prompt
- RL for Improving Reasoning
  - Test-time compute -> reasoning traces + RL

Example of few-shot prompts:

```
Solve each math problem carefully. Show the key steps briefly, then give the final a

Example 1:
Problem: ...
Solution:
Step 1: ...
Step 2: ...
Answer: ...

Now solve:
Problem: ...
Solution:
```

Template with reasoning

```
You are a math solver. Return only the final answer.

Example 1:
Problem: <problem>
Answer: <final answer>

Example 2:
Problem: <problem>
Answer: <final answer>

...

Now solve:
Problem: <new problem>
Answer:
```

Template without reasoning

UCLA ENGINEERING
Computer Science

Example of few-shot prompts:

```
Solve each math problem carefully. Show the key steps briefly, then give the final a

Example 1:
Problem: ...
Solution:
Step 1: ...
Step 2: ...
Answer: ...

Now solve:
Problem: ...
Solution:
```

Template with reasoning

Example (Fewshot + reasoning)

```
You are a careful math solver. For each problem, think through the solution step by

Example 1:
Problem: A notebook costs $3 and a pen costs $2. How much do 4 notebooks and 3 pens
Solution:
4 notebooks cost 4 × 3 = 12
3 pens cost 3 × 2 = 6
Total cost = 12 + 6 = 18
Answer: $18

Example 2:
Problem: Solve for x: 2x + 5 = 17
Solution:
Subtract 5 from both sides: 2x = 12
Divide both sides by 2: x = 6
Answer: x = 6

Example 3:
Problem: A rectangle has length 8 and width 5. What is its area?
Solution:
Area = length × width
Area = 8 × 5 = 40
Answer: 40

Now solve this:
Problem: A train travels 60 miles in 1.5 hours. What is its average speed?
Solution:
```

Example with reasoning

Example (Fewshot + reasoning)

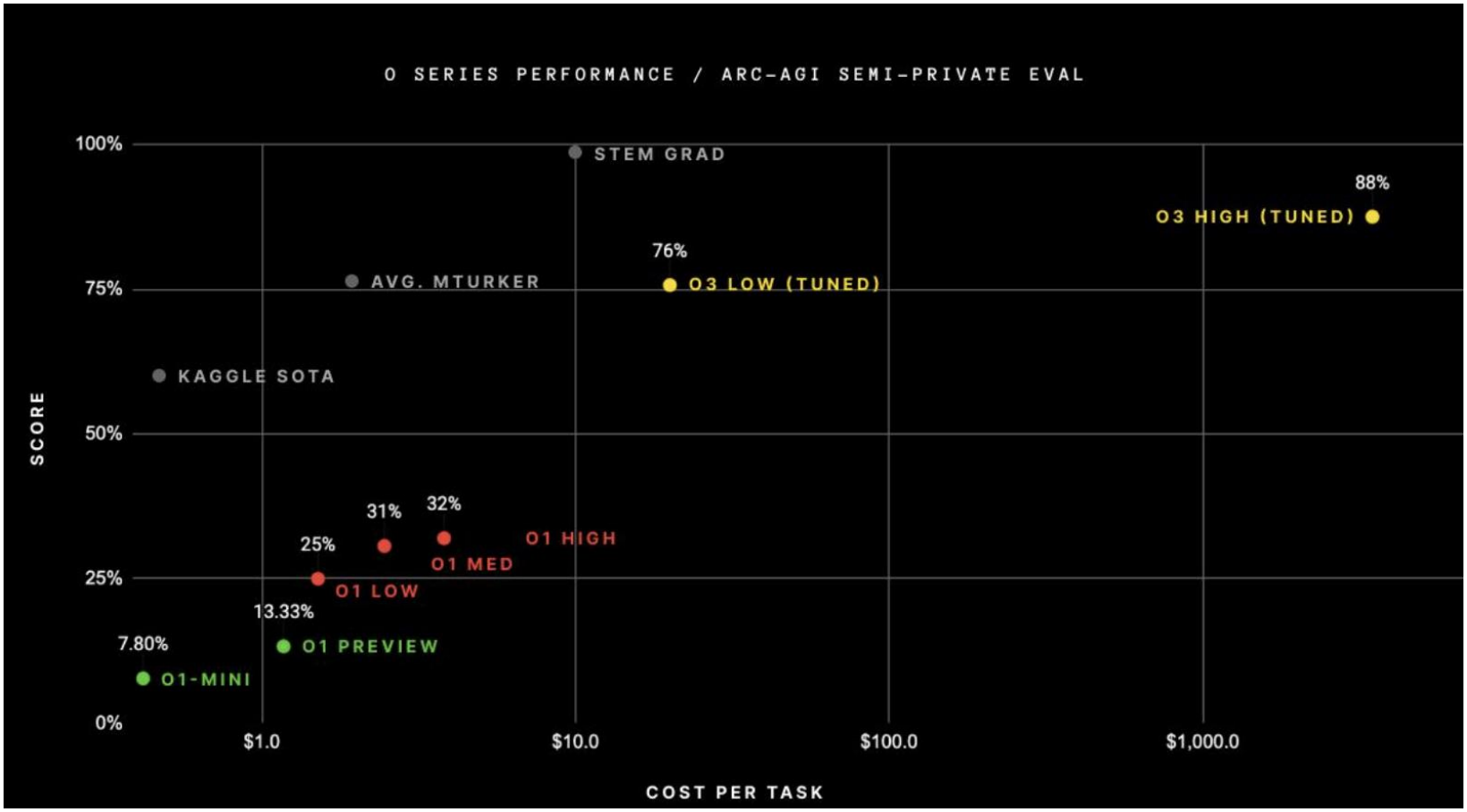
```
You are a careful math solver. For each problem, think through the solution step by

Example 1:
Problem: A notebook costs $3 and a pen costs $2. How much do 4 notebooks and 3 pens
Solution:
4 notebooks cost 4 × 3 = 12
3 pens cost 3 × 2 = 6
Total cost = 12 + 6 = 18
Answer: $18

Example 2:
Problem: So
Solution:
Subtract 5
Divide both
Answer: x =

Example 3:
Problem: A
Solution:
Area = leng
Area = 8 ×
Answer: 40

Now solve t
Problem: A
Solution:
```

Example with reasoning

```
You are a math solver. Return only the final answer. Do not show steps.

Example 1:
Problem: A notebook costs $3 and a pen costs $2. How much do 4 notebooks and 3 pens
Answer: $18

Example 2:
Problem: Solve for x: 2x + 5 = 17
Answer: x = 6

Example 3:
Problem: A rectangle has length 8 and width 5. What is its area?
Answer: 40

Now solve:
Problem: A train travels 60 miles in 1.5 hours. What is its average speed?
Answer:
```

Example without reasoning

Example of scaling test-time compute



https://arcprize.org/blog/oai-o3-pub-breakthrough